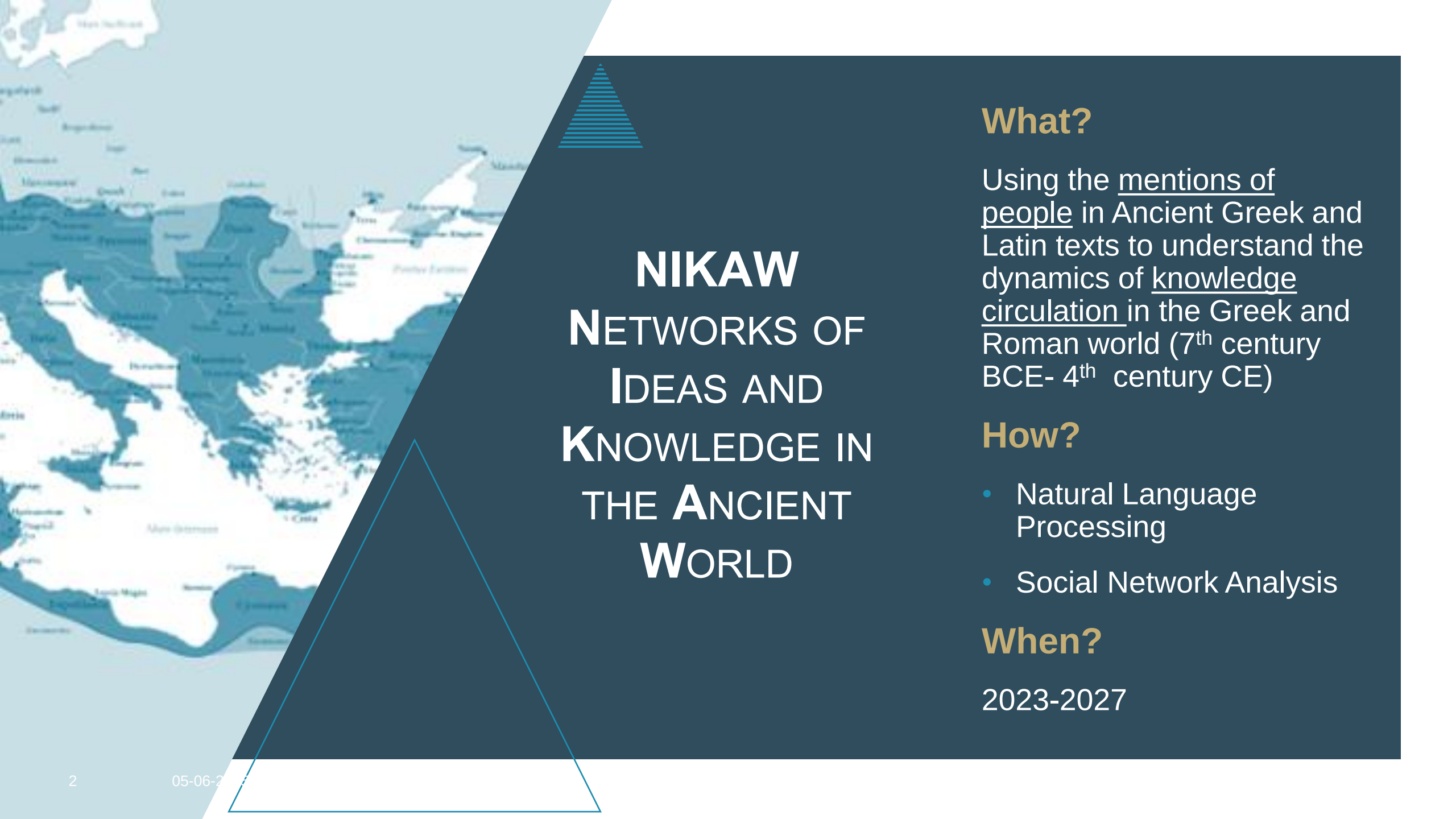


Intellectual Traditions in Ancient Greek (and Latin) texts: A Co-occurrence Network Approach

DH Benelux, 05-06-2025

Evelien de Graaf



NIKAW NETWORKS OF IDEAS AND KNOWLEDGE IN THE ANCIENT WORLD

What?

Using the mentions of people in Ancient Greek and Latin texts to understand the dynamics of knowledge circulation in the Greek and Roman world (7th century BCE- 4th century CE)

How?

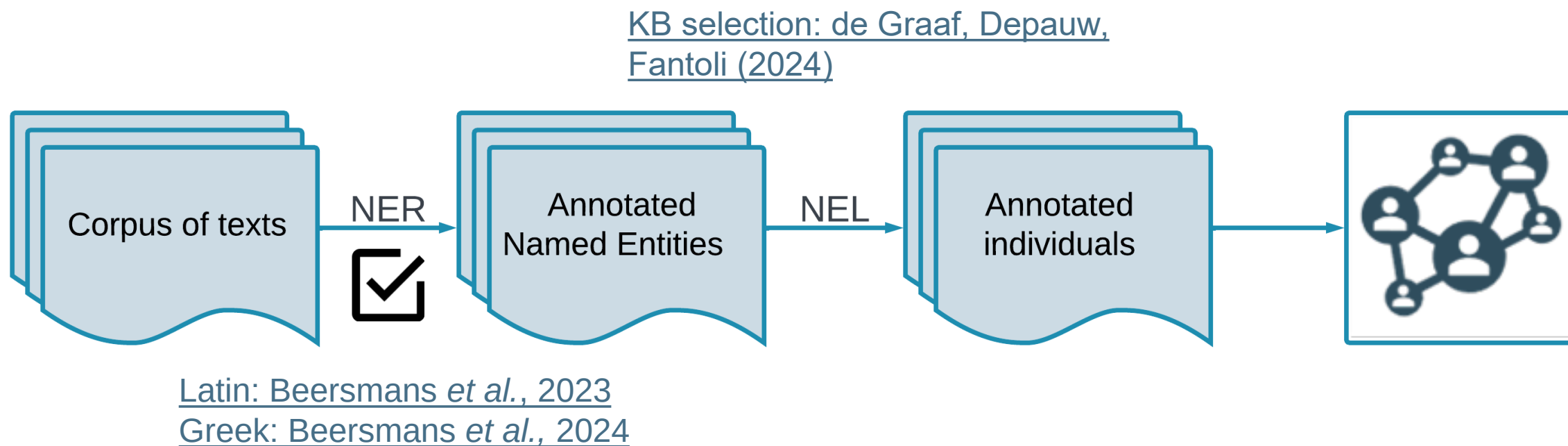
- Natural Language Processing
- Social Network Analysis

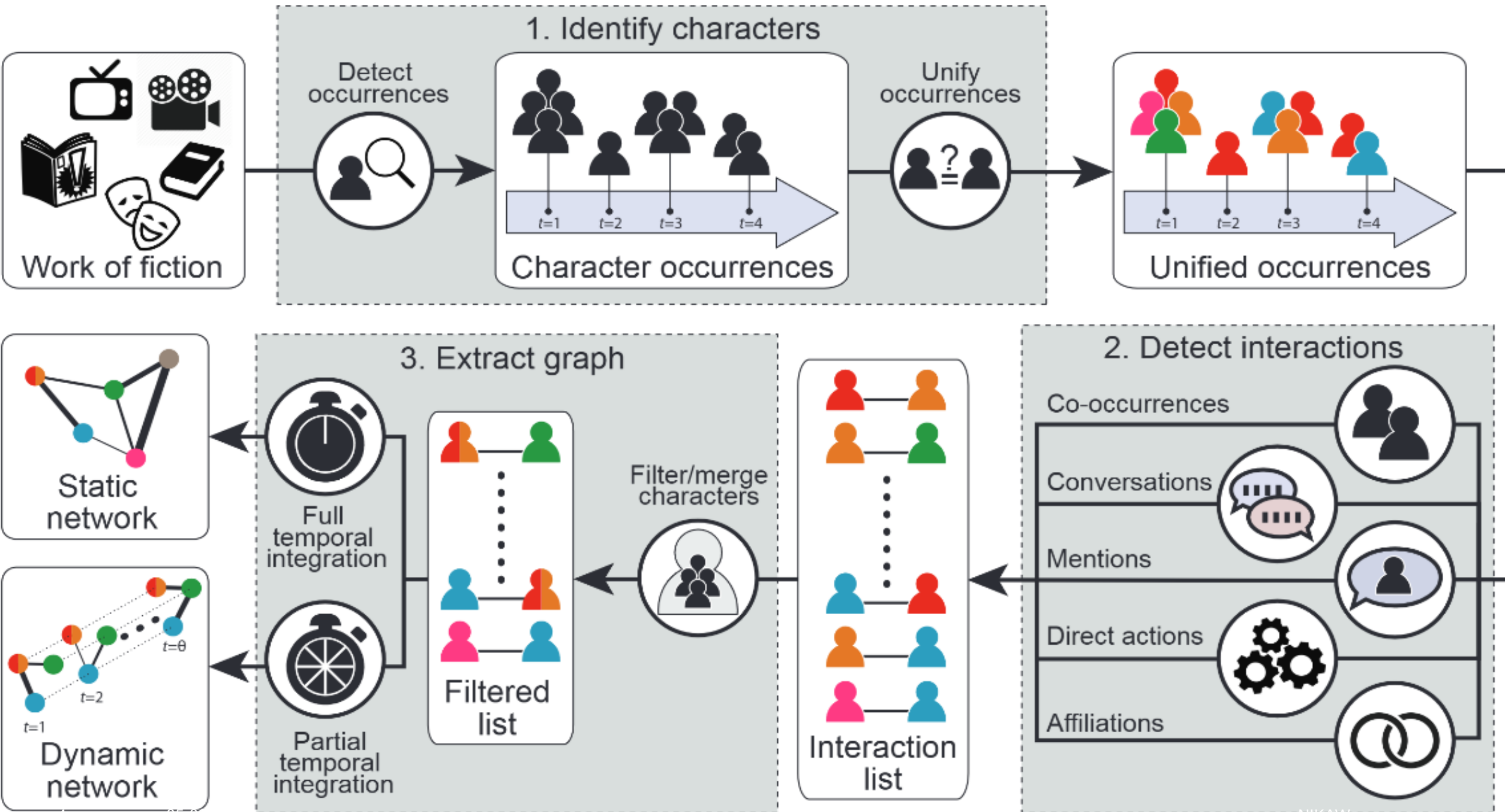
When?

2023-2027

OUR GOAL

- Automatically create networks based on co-occurrence in Greek and Latin literature to study co-occurrence of individuals in text





PRE-PROCESSING PAULYS REALENCYCLOPÄDIE

(COLLABORATION WITH THE TRISMEGISTOS+ TEAM)



- Local copy based on Register (98,577 articles)
 - Categorized based on keywords
 - Manual check
- 48,749 entries identified as Persons
 - Complete name restored & split into constituent elements (e.g. Caius Iulius Caesar = “Caius”, “Iulius”, “Caesar”)
 - Linked to TM name variant (TM NamVar), name (TM Nam) and author (TM Author)
 - Pageweight based on the *Seite* (pages)

A search tool facilitating access to the *RE* based on this work is now available at <https://www.trismegistos.org/real>

CASE STUDY: TEXTS WITH MENTIONS OF PLATO

Prolific author with well documented influence on both Christian and Non-Christian authors

Proof of concept NIKAW

- Clear picture of strengths and limitations of approach

Gold annotated NEL data

- Further evaluation of coverage selected Knowledge Base (KB): Paulys Realencyclopädie der classischen Altertumswissenschaft (RE)
- Can be used for training models

Network model

- Clear definition of Nodes, Edges, and Attributes
- Clear definition of network properties and appropriate analytics

SELECTED TEXTS: BASE CRITERIA

Lemma “Platon”/
“Πλάτων” > 5 times

Between 800 BCE
and 450 CE

Varying text types

Varying intellectual
backgrounds

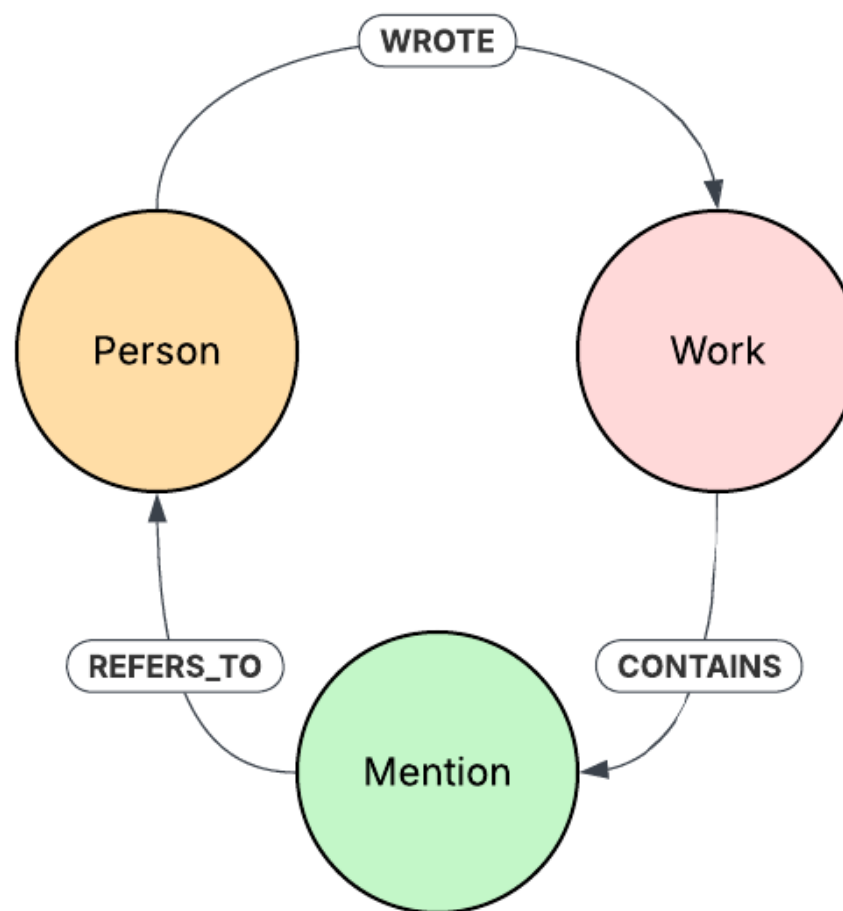
Author	Work	Period	Christian work
Greek [GLAUX corpus]			
[Plato]	Epistulae	BCE	☐
Aristoteles	Metaphysica	BCE	☐
Strabo	Geographica	BCE	☐
Dionysius Halicarnassensis	De Demosthenis dictione	BCE	☐
Plutarchus	Quaestiones convivales	CE	☐
Claudius Aelianus	Varia historia	CE	☐
Clemens Alexandrinus	Stromata	CE	☒
Origenes	Contra Celsum	CE	☒
Diogenes Laertius	Vitae philosophorum	CE	☐
Galenus	De placitis Hippocratis et Platonis	CE	☐
Latin [LASLA / Corpus Latin]			
Cicero	Tusculanae Disputationes	BCE	☐
Tertullian	De Anima	CE	☒
Seneca	Ad Lucilium Epistulae Morales	CE	☐
Lactantius	Divinarum Institutionum	CE	☒
Apuleius	Pro Se De Magia Liber	CE	☐



Social Network Analysis

MULTI-MODAL GRAPH SCHEMA

- Attributes
- ID
 - Complete name
 - Dates
 - Gender
 - Author
 - School of Thought
 - Pageweight
 - Total occurrences



- Attributes
- ID
 - Date
 - Language
 - Genre

- Attributes
- ID
 - Reference

CO-OCCURRENCE NETWORKS: INSPIRATION

Character constellations

Literary Characters

- Proximity
- Interaction

Bibliometrics

Authors in scholarly articles

- Proximity
- Similarity

Author co-mention analysis

Authors mentioned in reviews

- Proximity

Archeology

Material culture

- Similarity



Methodological question:
Can proximity be used to
approximate similarity between
individuals recorded in our
corpus? (Cf. Grayson et al.
2016; Smeets 2021)

For full bibliography on these topics, see the [abstract on Zenodo](#)

CO-OCCURRENCE PROJECTIONS

Co-occurrence (A,B) exists if person A co-occurs with person B within pre-defined sub-section (div_1, smallest div and sliding window of two sentences)

Weighted

The aggregated weight for two entities

$$w(A, B) = \frac{N}{\sum(\frac{1}{d_i})}$$

where:

- N = number of distance measurements (total co-occurring mention pairs)
- d_i = distance between individual mentions of A and B



First experiments: Greek corpus

NETWORK SIZE

		Div_1	Div_small	Sliding window
Nodes		3.041	3001	2921
Edges	Unweighted	615.423	151.398	19.777
	Weighted	568.155	145.790	18.295

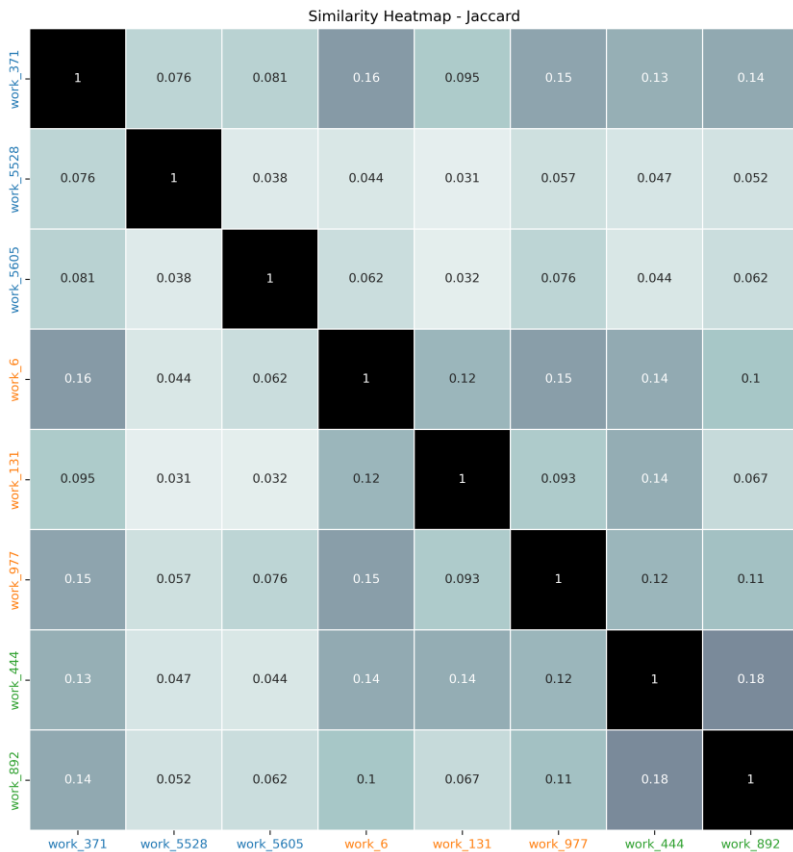
NODES IN MULTIPLE DISTINCT WORKS

		Div_1	Div_small	Sliding window
Nodes		715	713	713
Edges	Unweighted	164.565 (26.74%)	43.590 (28.79%)	8.610 (43.54%)
	Weighted	117.297 (20.65%)	37.982 (26.05%)	7.128 (38.96%)

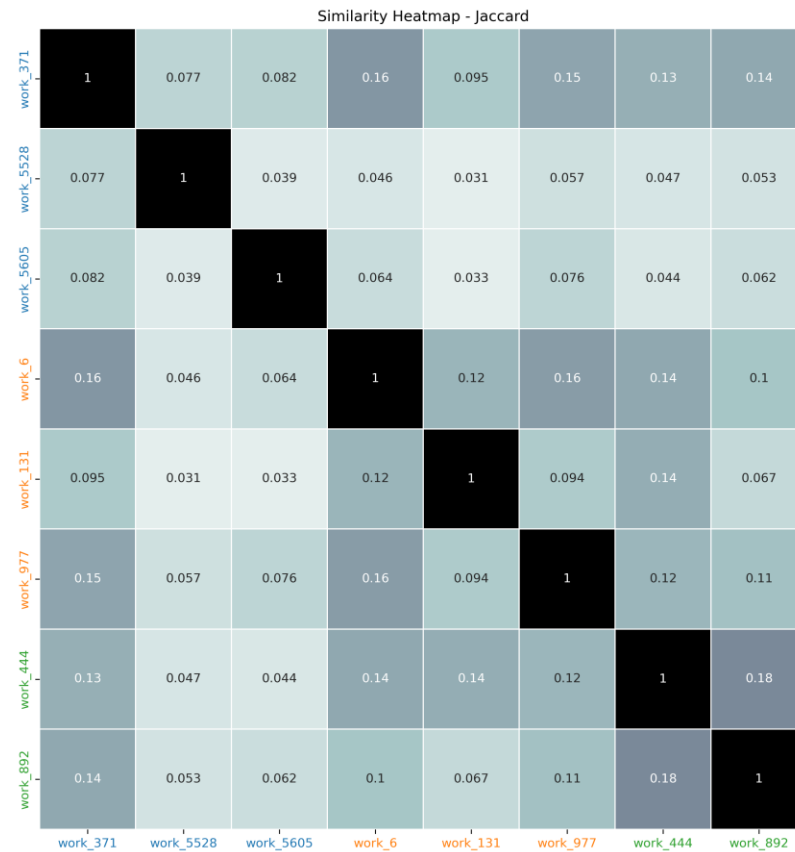
JACCARD SIMILARITY – NODES PER WORK



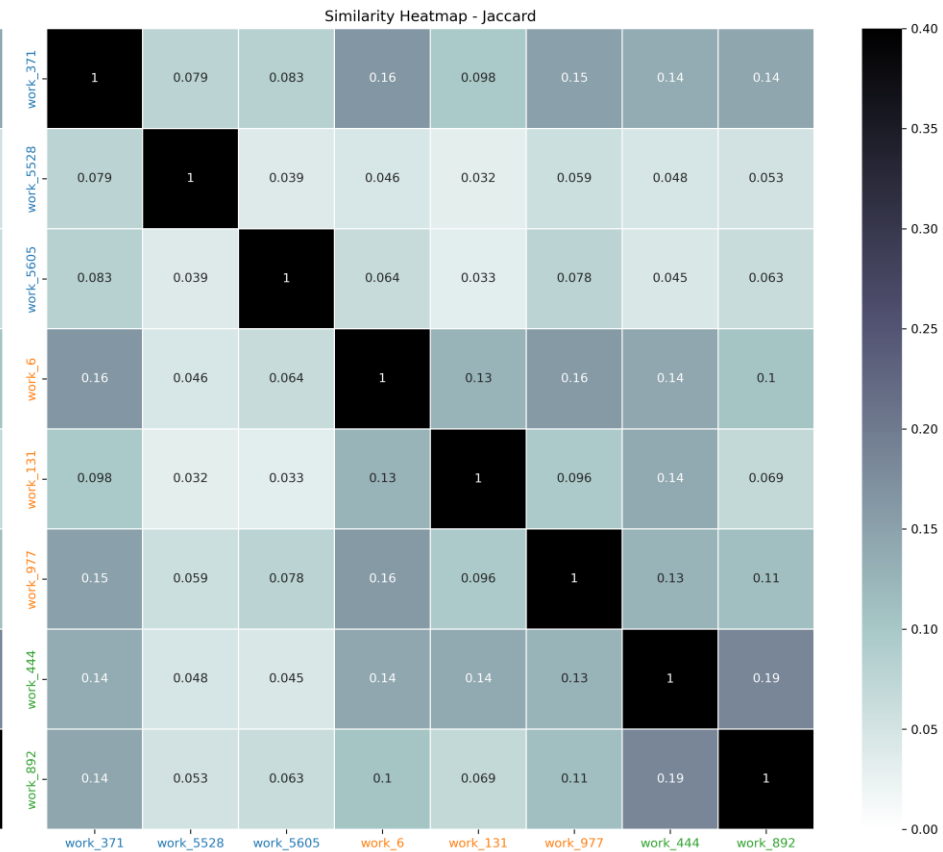
Div_1



Div_small



Sliding window



Groups

- BCE Non-Christian
- CE Non-Christian
- CE Christian



CENTRAL QUESTION

Can proximity be used to detect meaningful clustering as is done in other applications of SNA, such as ACA and CPA, in our corpus?

→ Remains open for future evaluation...



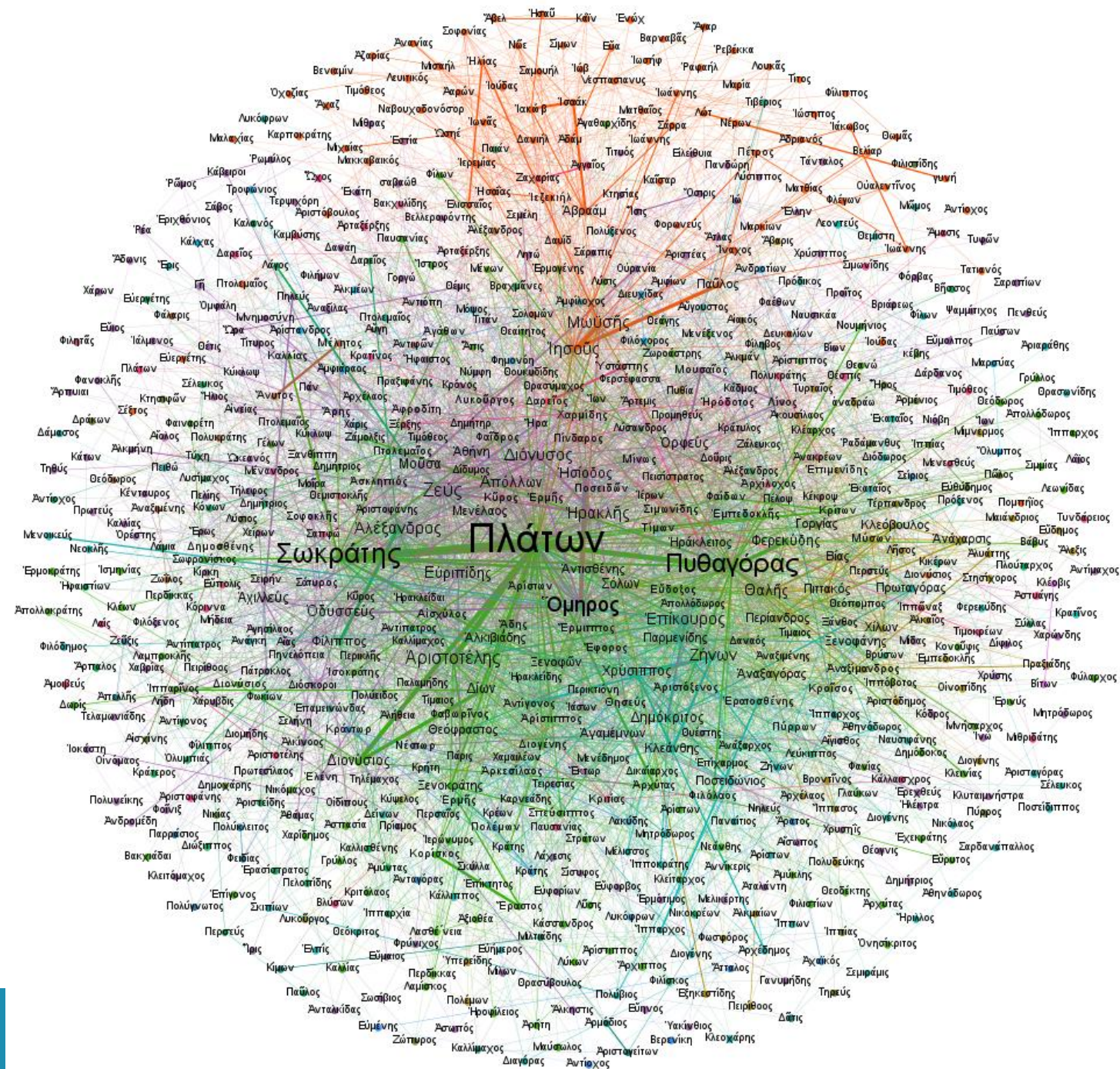
SLIDING WINDOW

Weighted edges

Fruchterman Reingold

Colour: Modularity (weighted)

Size: Degree



SLIDING WINDOW

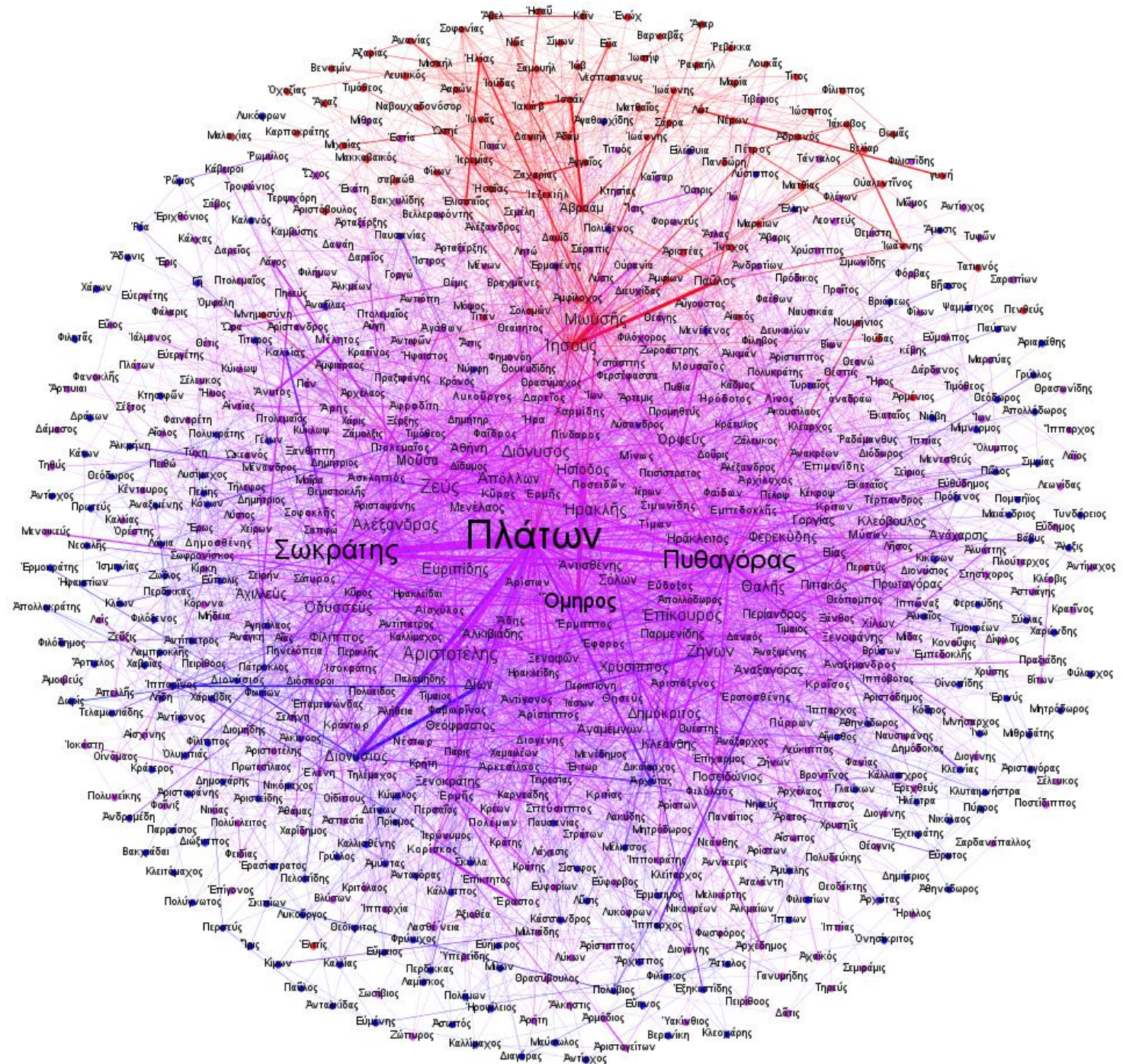
Weighted edges

Fruchterman Reingold

Colour:

■	Christian_mixed	(61,15%)
■	Non_Christian_only	(26,37%)
■	Christian_only	(12,48%)

Size: Degree

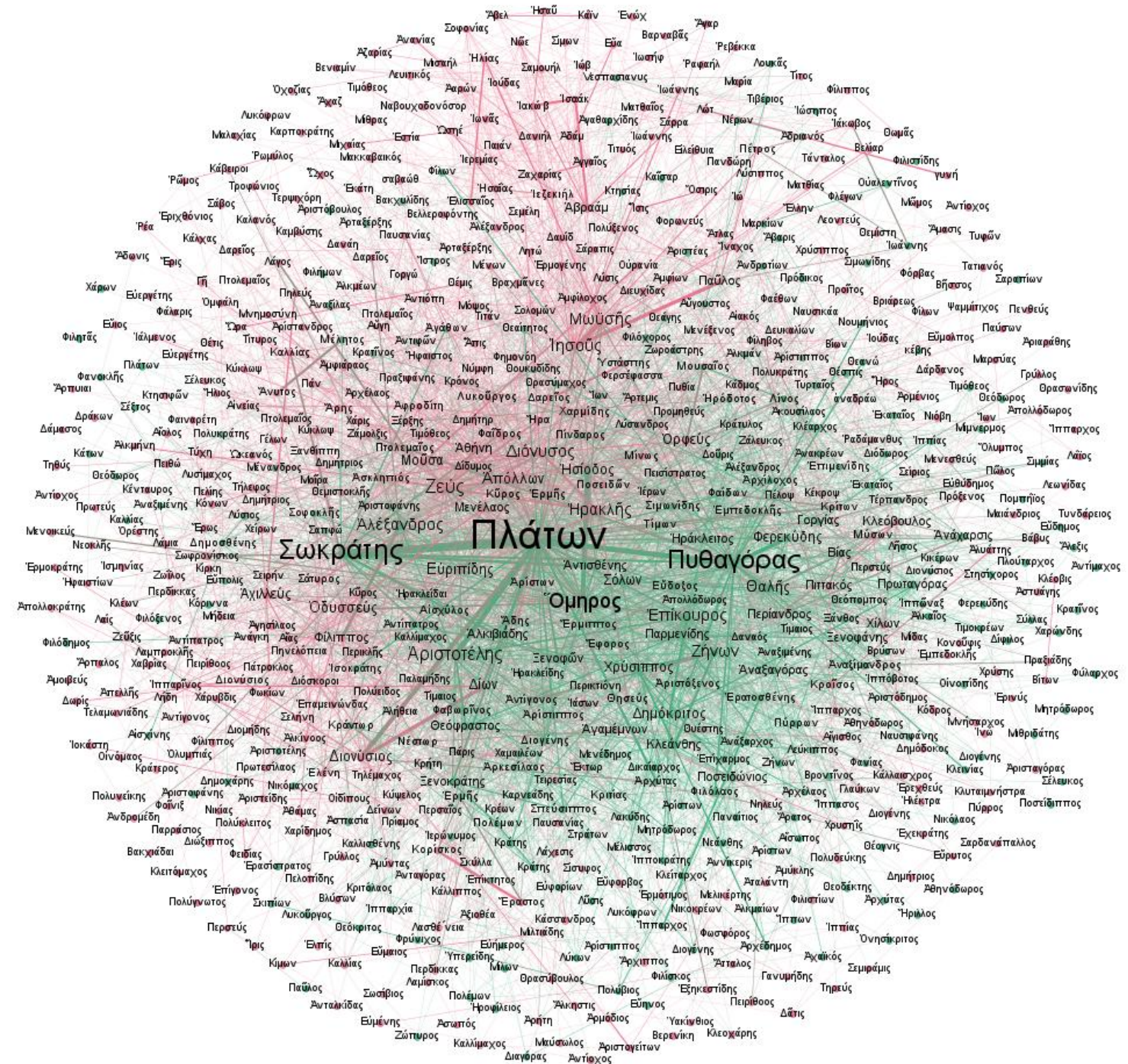


SLIDING WINDOW

Weighted edges

Fruchterman Reingold

Colour: Author



Size: Degree

FUTURE PLANS AND POTENTIAL

- Structurally record **attributes** and study the influence of attributes on the structure of the network
- Investigate whether shared patterns in mentions between authors relate to any part of the background of the author (ie. their education, their social group)
- Include mention context in study
 - Epithets, patronymics, ethnics etc.
 - Sentiment of context: positive vs. negative

THANK YOU!